ClaimVerify: An AI-Powered Hybrid Fact Verification System Combining Semantic Retrieval, Transformer Classification and Web-Augmented Evidence

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***Abstract*—The widespread circulation of unverified information across digital platforms has intensified the societal need for scalable, reliable and transparent fact-verification tools. This report presents *ClaimVerify*, an AI-powered hybrid fact-verification system designed to assess the credibility of natural-language claims by integrating offline semantic retrieval, transformer-based classification, calibrated uncertainty estimation, and web-augmented evidence retrieval powered by Google Custom Search API. The system combines a high-recall FAISS index built from major fact-checking sources with MiniLM sentence embeddings, achieving near-perfect Recall@5 (1.0000) for offline evidence retrieval. A fine-tuned RoBERTa classifier with temperature scaling delivers balanced multi-class performance (Macro F1 = 0.5057; Brier score = 0.1989), particularly improving fairness across imbalanced classes. When offline evidence coverage is weak, ClaimVerify triggers a Google Custom Search-based fallback, enabling dynamic external evidence acquisition. The system further incorporates Integrated Gradients to provide token-level explanations, enhancing Interpretability and User trust. A production-ready Streamlit interface unifies evidence visualization, verdict interpretation, uncertainty communication and runtime profiling into an accessible user experience. This report details the full lifecycle of ClaimVerify—from data engineering and model development to evaluation and responsible AI considerations—positioning it as a practical and extensible framework for real-world misinformation detection.**

***Index Terms*—Fact verification, Semantic retrieval, RoBERTa, Transformer models, Misinformation detection, Google custom search API, Responsible AI**

I. Introduction

The rapid acceleration of misinformation across digital platforms has emerged as a critical societal and technological challenge. Social media ecosystems increasingly amplify unverified claims, politically motivated narratives and misleading statements that can influence public opinion, distort policy discussions and erode trust in institutions. Traditional fact-checking organizations such as PolitiFact and Snopes provide high-quality verification, but their processes are inherently labor-intensive and cannot scale to the volume or velocity of modern information flows. This gap underscores the need for automated, transparent and trustworthy AI-driven systems capable of assisting users in evaluating the credibility of natural-language claims.

Recent advances in natural language processing (NLP), particularly large-scale transformer models [1]–[3], have substantially improved the ability of machines to understand and reason over text. In parallel, vector similarity search tools such as FAISS have enabled efficient evidence retrieval from large corpora [4]. However, despite these advances, fact verification remains a complex, multi-stage problem requiring the integration of semantic retrieval, classification, calibration, and interpretability along with responsible AI design. Many existing verification datasets, including LIAR [6] and FEVER [7], demonstrate the feasibility of automated fact-checking research, yet real-world systems must overcome limitations such as data imbalance, semantic ambiguity and limited coverage for emerging or previously unseen claims.

A. Problem Statement

ClaimVerify addresses the central challenge of determining whether a natural-language claim is likely to be true, false, or uncertain by grounding predictions in evidence retrieved from verified fact-checking databases. A key difficulty lies in covering the long-tail of novel claims that lack close matches within offline datasets. Systems that rely solely on classification or retrieval struggle in such settings, increasing the risk of misleading predictions. Therefore, the core problem is to design a system that not only performs well on known fact-checks but also remains reliable, interpretable and adaptive when confronted with unfamiliar or emerging information.

B. Project Objectives

The primary objective of this project is to develop an end-to-end AI-powered fact verification system capable of:

- Retrieving semantically relevant evidence using high-recall sentence embeddings and FAISS similarity search;
- Classifying claims into *Likely True*, *Likely False*, or *Uncertain* using a fine-tuned RoBERTa model [1];

- Producing calibrated confidence scores using neural network temperature scaling [5];
- Handling unseen or poorly matched claims via a web-augmented retrieval fallback powered by Google Custom Search [11];
- Providing interpretable token-level explanations using Integrated Gradients;
- Presenting an intuitive, user-centered interface that communicates uncertainty responsibly.

Together, these components form a multi-stage pipeline designed to support human decision-making rather than replace expert judgment.

C. Scope and Contributions

This work makes several contributions toward building a practical and extensible fact-verification system:

- **Hybrid Retrieval Architecture**: A novel two-tier retrieval mechanism that combines offline FAISS search with web-based fallback retrieval to provide robust coverage for both known and emerging claims.
- **Calibrated Transformer Classification**: A fine-tuned RoBERTa classifier augmented with temperature scaling to produce well-calibrated probability estimates, addressing overconfidence commonly observed in transformer-based models.
- **Evidence-Aware Transparency**: Integrated Gradients explanations and structured evidence visualization that enhance interpretability and help users understand the basis of model predictions.
- **Unified Data Pipeline**: Construction of a consolidated fact-checking dataset from Snopes and PolitiFact [13], [14], including preprocessing, normalization, and label harmonization.
- **Deployment-Ready Interface**: A production-grade Streamlit application [15] designed to make AI-driven verification accessible to non-technical users through a clear, interpretable interface.

These contributions collectively advance the development of practical, trustworthy and transparent AI systems for misinformation mitigation.

D. Report Organization

The remainder of this report is structured as follows. Section II reviews related work across fact-checking datasets, retrieval-based verification, and transformer-based misinformation detection. Section III describes the system design and implementation, including data engineering, retrieval architecture, classification pipeline, calibration strategy, explainability methods, and UI considerations. Section IV presents evaluation results for retrieval quality, classification performance, calibration behavior, and end-to-end inference. Section V discusses system strengths, limitations, and broader implications. Section VI outlines future work directions, followed by a Responsible AI analysis in Section VII. Section VIII concludes with final remarks on the system's contributions.

II. Related Work

Fact verification has emerged as a critical research area as misinformation continues to spread rapidly across digital platforms. Early approaches to automated fact-checking focused primarily on detecting deceptive writing styles or shallow linguistic cues, but these models often failed to incorporate external evidence or contextual reasoning. Recent advances in Natural language processing (NLP) and retrieval-augmented architectures have led to more robust systems capable of grounding predictions in factual context.

One major line of work has centered on Transformer-based Language models, which introduced contextualized representations through self-attention mechanisms. The foundational transformer architecture proposed by Vaswani et al. [3] enabled large-scale pretraining and laid the groundwork for modern fact-verification systems. RoBERTa [1], an optimized and more robust variant of BERT, improved training stability and downstream task performance, making it widely adopted in text classification pipelines, including misinformation detection. Complementary embedding models such as Sentence-BERT (SBERT) [2] demonstrated that transformers could generate high-quality sentence embeddings suitable for semantic similarity, greatly benefiting evidence retrieval tasks.

Several benchmark datasets have also shaped progress in the field. The LIAR dataset [6] introduced one of the earliest large-scale collections of political claims paired with fact-check labels. FEVER [7] expanded the paradigm by requiring systems not only to classify claims but also to retrieve supporting evidence from Wikipedia, pushing research toward retrieval-integrated architectures. Additional work by Hanselowski et al. [8] and Shaar et al. [9] further explored stance detection and previously fact-checked claim retrieval, highlighting the importance of grounding model predictions in factual databases rather than relying solely on linguistic features.

Increasing attention has been given to evidence-aware neural models. Popat et al. [10] proposed DeClarE, which integrates evidence sentences with attention-based reasoning to detect false claims. These models demonstrated the value of coupling retrieval with classification, but they often relied on limited or domain-specific corpora, reducing generality. More recently, vector search libraries such as FAISS [4] have enabled efficient large-scale semantic retrieval, making it feasible for fact-checking systems to search tens of thousands of verified claims in real time.

Despite progress, several gaps remain in prior research. Many systems assume access to large, clean and domain-consistent datasets, which is rarely the case in real-world misinformation scenarios. Dataset imbalance, Varied linguistic framing of claims and Unstable classifier confidence continue to challenge system reliability. Moreover, most existing works lack integrated fallback strategies when retrieval fails, and few incorporate probability calibration techniques such as those proposed by Guo et al. [5], which are critical for trustworthy deployment.

ClaimVerify advances the state of the art in three key ways:

1) **Hybrid Retrieval:** It combines offline semantic retrieval with an online fallback mechanism using the Google Custom Search API [11], enabling coverage of both previously fact-checked claims and emerging real-time claims.
2) **Calibrated Classification:** It integrates calibrated RoBERTa classification with temperature scaling, addressing the confidence miscalibration issues identified in earlier work [5].
3) **Token-Level Interpretability:** It incorporates token-level interpretability using Integrated Gradients, providing transparent explanations for classifier predictions which is an area where many prior systems offer limited visibility.

Overall, ClaimVerify builds upon and extends past research by unifying evidence retrieval, calibrated classification, real-time fallback search, and explainability into a single operational pipeline, bringing research prototypes closer to a deployable, user-facing fact-verification system.

III. System Design and Implementation

A. System Overview

ClaimVerify is designed as a modular, end-to-end AI-powered fact verification system that integrates semantic retrieval, calibrated classification and interpretable reasoning within a single operational pipeline. The system architecture, illustrated in Figure 1, follows a layered design that cleanly separates user interaction, retrieval logic, inference and explainability. This architectural organization enables robustness, scalability and reproducibility while supporting real-time claim verification in practical settings.

At a high level, ClaimVerify transforms a free-form natural-language claim into a structured, evidence-grounded verdict through a sequence of well-defined computational stages. The system emphasizes *evidence-first reasoning*, ensuring that predictions are contextualized using retrieved fact-checks rather than relying solely on linguistic cues. This design choice aligns the system more closely with real-world fact-checking workflows used by professional organizations.

The pipeline begins with an **Input Layer**, where users submit claims through a web-based interface. This layer performs lightweight validation to ensure compatibility with downstream transformer models, including input length constraints and encoding checks. The goal at this stage is not semantic interpretation, but reliable ingestion of diverse user-generated text.

The claim then enters the **Preprocessing and Retrieval Engine**. Deterministic text normalization is applied to reduce superficial variation across claims, including lowercasing, punctuation normalization, unicode standardization, and whitespace cleanup. The normalized text is embedded using a MiniLM-based SentenceTransformer model, which produces dense semantic representations optimized for similarity search [2]. These embeddings are queried against a FAISS index containing approximately 25,000 verified claims sourced from PolitiFact and Snopes [4]. FAISS enables efficient large-scale nearest-neighbor retrieval, allowing ClaimVerify to identify semantically similar fact-checks with low latency.

A central architectural contribution of ClaimVerify lies in its **Hybrid Inference Layer**. After offline retrieval, the system evaluates the similarity score of the top-ranked evidence. When this score exceeds a predefined threshold (0.82 in the final system), the offline evidence is considered sufficiently reliable and is used directly. If the similarity falls below this threshold, the system activates an online fallback mechanism powered by the Google Custom Search JSON API [11]. This fallback retrieves real-time external evidence, enabling the system to handle emerging or previously unverified claims that are poorly represented in the offline database. The hybrid strategy balances precision and coverage while explicitly tracking evidence provenance for user transparency.

In parallel with retrieval, the claim is processed by the **Model Layer**, which performs claim-level classification. ClaimVerify employs a fine-tuned RoBERTa-based transformer model [1] trained to predict three classes: *Likely True*, *Likely False*, and *Uncertain*. To address the well-documented issue of overconfident neural predictions, the system applies temperature scaling to calibrate output probabilities [5]. The final model uses a learned temperature of $T = 1.1678$, producing confidence scores that better reflect empirical accuracy and uncertainty.

To improve transparency and user trust, ClaimVerify integrates **Token-level explainability** using Integrated Gradients [8]. This method attributes the classifier's decision to individual input tokens, allowing users to inspect which parts of a claim most strongly influenced the prediction. Explainability is treated as a core system output rather than a post-hoc diagnostic feature.

Finally, results are presented through the **Interface Layer**, implemented using Streamlit [15]. The interface displays the predicted verdict, calibrated confidence score, ranked evidence cards with source links, token-level importance heatmaps, and runtime profiling information. Importantly, the interface communicates uncertainty explicitly by flagging low-confidence predictions and indicating whether evidence originated from the offline database or online fallback.

Overall, the architecture shown in Figure 1 reflects a hybrid, evidence-centered approach to automated fact verification. By unifying semantic retrieval, calibrated transformer-based classification, real-time fallback search, and interpretable outputs within a single system, ClaimVerify advances beyond prior research prototypes toward a deployable, user-facing fact verification platform.

B. Lifecycle Stages

This section details the end-to-end lifecycle of the ClaimVerify system, covering data collection and preprocessing, model development and evaluation methodology. Emphasis is placed on design decisions, challenges encountered, and the rationale behind architectural and modeling choices,

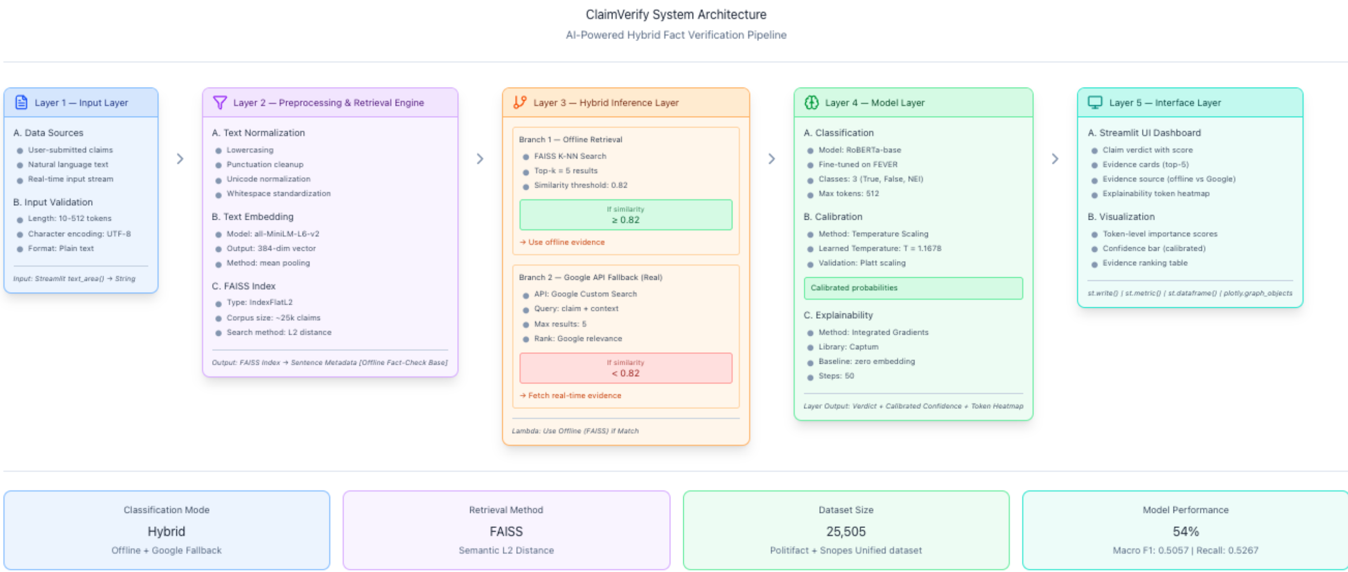

Fig. 1. ClaimVerify system architecture illustrating the end-to-end hybrid fact verification pipeline, including offline FAISS retrieval, Google Search API fallback, calibrated RoBERTa classification, and token-level explainability.

reflecting both academic rigor and real-world deployment considerations.

C. Data Collection and Preprocessing

1) Data Sources: ClaimVerify relies exclusively on publicly available, professionally curated fact-checking datasets to ensure transparency, credibility and reproducibility. Two primary data sources were used and subsequently unified into a single dataset:

- **Snopes Fact-News Dataset** [13]
- **PolitiFact Fact-Check Dataset** [14]

These datasets were selected because they represent authoritative fact-checking organizations with long-standing editorial standards and cover a wide range of political, social, scientific, and health-related claims.

Snopes Dataset loaded: (4550, 9)

	question	comment	claim	rate	what's true	what's false	what's unknown	origin	summary
0	Did Kamala Harris Support Abortion Until the T...	The Democratic vice-presidential candidate's p...	U.S. Sen. Kamala Harris, D-Calif., supports th...	Mixture	U.S. Sen. Kamala Harris, D-Calif, has maintain...	She has not explicitly stated that she support...	NaN	Debates surrounding abortion frequently resurf...	U.S. Sen. Kamala Harris, D-Calif., has been to...
1	Did Hitler Invent the Inflatable Sex Doll?	Nothing attracts curiosity like a lurid combin...	Adolf Hitler was behind the invention of the f...	False	NaN	NaN	NaN	In late July 2020, readers shared a 2016 blog ...	Readers shared a 2016 blog post with the headl...
2	Does Texting 'RBG' to 50409 Send a Letter to Y...	The automated political advocacy service Resis...	Texting "RBG" to 50409 engages a service calle...	True	NaN	NaN	NaN	In September 2020, following the death of U.S....	Readers asked Snopes to examine the accuracy a...

Fig. 2. Overview of the Snopes fact-checking dataset, illustrating claim categories and verdict distribution.

a) Snopes Dataset.: The Snopes dataset contains fact-checked claims spanning misinformation, hoaxes, and viral narratives. It includes diverse verdict labels such as *True*, *False*, *Mixture*, *Unproven*, and *Miscaptioned*. While rich in coverage, this dataset introduces challenges related to label ambiguity and heterogeneous verdict semantics.

PolitiFact Dataset loaded: (21152, 8)

	verdict	statement_originator	statement	statement_date	statement_source	factchecker	factcheck_date	factcheck_analy
0	true	Barack Obama	John McCain opposed bankruptcy protections for...	6/11/2008	speech	Adriel Bettelheim	6/16/2008	https://www.politifact.com/factchecks/20
1	false	Matt Gaetz	"Bennie Thompson actively cheer-led riots in t...	6/7/2022	television	Yacob Reyes	6/13/2022	https://www.politifact.com/factchecks/20
2	mostly-true	Kelly Ayotte	Says Maggie Hassan was "out of state on 30 day...	5/18/2016	news	Clay Wirestone	5/27/2016	https://www.politifact.com/factchecks/20
3	false	Bloggers	"BUSTED: CDC Inflated COVID Numbers, Accused o...	2/1/2021	blog	Madison Czopek	2/5/2021	https://www.politifact.com/factchecks/20
4	half-true	Bobby Jindal	"I'm the only (Republican) candidate that has ...	8/30/2015	television	Linda Qiu	8/30/2015	https://www.politifact.com/factchecks/20
5	true	S.E. Cupp	"There are actually only 30 countries that pra...	8/23/2015	news	Will Cabaniss	8/23/2015	https://www.politifact.com/factchecks/20

Fig. 3. Overview of the PolitiFact fact-checking dataset, including label distribution and claim frequency.

b) PolitiFact Dataset.: The PolitiFact dataset primarily focuses on political claims and public statements, providing verdicts such as *True*, *Mostly True*, *Half True*, *Mostly False*, and *False*. Compared to Snopes, PolitiFact offers more consistent labeling but is more domain-constrained.

2) Unified Dataset Construction: To enable a single retrieval and classification pipeline, both datasets were merged into a unified corpus. This process required careful alignment of schema, labels, and textual content.

Key unification steps included:

- Schema harmonization across datasets (claim text, verdict, summary, source URL)
- Deduplication of semantically identical or near-identical claims
- Removal of incomplete or malformed entries
- Standardization of metadata fields

After cleaning and deduplication, the final unified dataset contained **25,505 unique claims**.

a) Label Mapping Strategy.: Given the heterogeneity of original verdicts, all labels were mapped into a three-class schema:

claim_id	claim_text	verdict_original	verdict_mapped	summary	originator	url	dataset_source
P02167	Speedway is offering 500$fuelcardsfor$ 1.95...	pants-fire	Likely False	NaN	Facebook posts	https://www.politifact.com/factchecks/2022/jun...	PolitiFact
P16561	"The federal government borrows $4 billion eve...	mostly-true	Likely True	NaN	National Republican Congressional Committee	https://www.politifact.com/factchecks/2011/may...	PolitiFact
P05571	"Yale study: Vaccinated people more likely to ...	false	Likely False	NaN	Instagram posts	https://www.politifact.com/factchecks/2022/jan...	PolitiFact
P04220	"The people that went to school	pants-fire	Likely False	NaN	Donald Trump	https://www.politifact.com/factchecks/2011/feb...	PolitiFact

Fig. 4. Unified dataset label distribution after preprocessing and deduplication.

- **Likely True**
- **Likely False**
- **Uncertain**

Ambiguous verdicts such as *Mixture*, *Half True*, and *Unproven* were intentionally mapped to *Uncertain* to avoid forcing overly confident predictions in borderline cases. This design choice directly supports responsible AI principles by prioritizing caution over false certainty.

3) Text Preprocessing Pipeline: All textual inputs pass through a unified preprocessing function applied consistently across retrieval, classification, and evaluation stages. The preprocessing pipeline includes:

- Lowercasing of all text
- Unicode normalization
- Removal of punctuation and special characters
- Whitespace normalization
- Standardized handling of quotes and hyphenation

This strict normalization reduces semantic drift between retrieval embeddings and classifier inputs, improving alignment between system components.

a) Challenges Encountered.: Key challenges during data preparation included:

- Significant class imbalance (majority *Likely False*)
- Inconsistent verdict granularity across datasets
- Variability in claim phrasing and narrative framing
- Limited representation of genuinely uncertain claims

These challenges motivated downstream modeling and calibration decisions discussed in the next subsection.

D. Model Development and Evaluation

1) Model Selection Rationale: ClaimVerify adopts a hybrid retrieval-classification architecture combining semantic search with supervised classification. Each model component was selected based on performance, efficiency and suitability for real-time deployment.

a) Semantic Retrieval Model.: For evidence retrieval, the system uses the **MiniLM-L6-v2** sentence transformer [2]. This model provides:

- High-quality sentence embeddings
- Low computational overhead
- Strong performance on semantic similarity tasks

Embeddings are indexed using FAISS [4], enabling efficient nearest-neighbor search over more than 25,000 claims with sub-second latency.

b) Classification Model.: Claim classification is performed using a fine-tuned **RoBERTa-base** transformer [1]. RoBERTa was selected due to:

- Strong empirical performance on natural language understanding tasks
- Robust pretraining using dynamic masking
- Proven effectiveness in misinformation detection settings

The classifier predicts one of three labels: *Likely True*, *Likely False*, or *Uncertain*.

2) Training Strategy: The unified dataset was split into training, validation, and test sets using a stratified strategy to preserve label proportions:

- Training set: 20,404 samples
- Validation set: 2,550 samples
- Test set: 2,551 samples

To address class imbalance, class-weighted loss was applied during training. Additional stability measures included:

- Gradient clipping
- Learning-rate warmup
- Early stopping based on validation loss
- Deterministic seed initialization

3) Evaluation Metrics: Model performance was evaluated using multiple complementary metrics:

- Accuracy
- Macro-averaged Precision, Recall, and F1-score
- Confusion matrix analysis
- Brier score for probability calibration

The final classifier achieved an overall accuracy of **54%**, with a macro F1-score of **0.5057**. While accuracy is moderate, macro-averaged metrics provide a more reliable assessment given dataset imbalance.

4) Calibration and Reliability: To improve confidence reliability, Temperature scaling [5] was applied post-training. The learned temperature parameter **T = 1.1678** significantly reduced overconfident predictions and improved the Brier score to **0.1989**.

This calibration step is critical for a user-facing system, ensuring that predicted confidence values meaningfully reflect empirical correctness.

Key Takeaway : Rather than optimizing solely for raw accuracy, ClaimVerify prioritizes balanced performance, calibrated confidence and interpretability—qualities essential for real-world fact verification systems.

5) Hybrid Retrieval with Google Custom Search API: While offline semantic retrieval provides high accuracy for previously fact-checked claims, real-world misinformation evolves rapidly and often includes novel or emerging narratives not present in static datasets. To address this limitation, ClaimVerify integrates a real-time online fallback mechanism using the Google Custom Search JSON API [11]. This hybrid design ensures that the system remains robust even when offline evidence coverage is insufficient.

a) Triggering Mechanism.: The Google Search fallback is activated conditionally based on retrieval confidence. After offline FAISS-based retrieval, the system evaluates the cosine

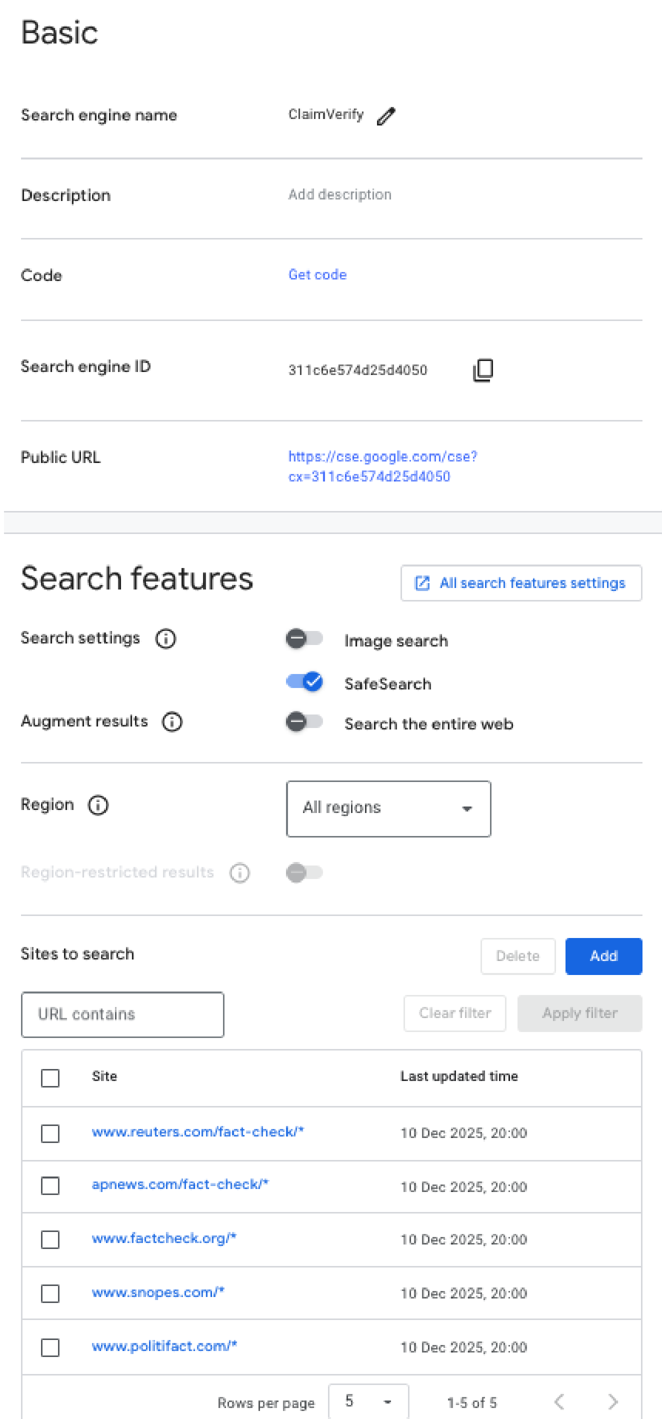

Fig. 5. Google Custom Search Engine configuration used for ClaimVerify, restricted to verified fact-checking domains.

similarity score of the top-ranked evidence item. If the similarity score falls below a predefined threshold (0.82), the claim is deemed insufficiently supported by the offline database, triggering the online retrieval pathway.

b) Search Engine Configuration.: The Google Custom Search Engine (CSE) was explicitly configured to prioritize authoritative fact-checking sources rather than the open web. As shown in Fig. 5, the search engine is restricted to the following verified domains:

- `reuters.com/fact-check`
- `apnews.com/fact-check`
- `factcheck.org`
- `snopes.com`
- `politifact.com`

This domain-restricted configuration significantly reduces the risk of retrieving unreliable or misleading content and ensures that external evidence aligns with professional journalistic standards.

c) Query and Ranking Strategy.: When triggered, the system submits the user's original claim text as a natural-language query to the Google Custom Search API. The API returns up to five ranked results based on Google's relevance scoring. Each result includes:

- Headline or title
- Snippet summary
- Source URL
- Domain metadata

Unlike offline retrieval, similarity scores are not computed for Google results. Instead, relevance is inferred from Google's ranking and the credibility of the source domain.

d) Integration with the Inference Pipeline.: Google-based evidence is seamlessly integrated into the same downstream inference flow as offline evidence. Retrieved results are presented uniformly in the interface as evidence cards, clearly labeled as *External Evidence* and distinguished from offline database matches. If the Google API fails to return results due to network or quota limitations, the system gracefully falls back to offline evidence and surfaces a transparency warning to the user.

e) Design Implications.: This hybrid retrieval strategy offers several advantages:

- Extends coverage to emerging and previously unseen claims
- Preserves reliability through domain restriction
- Enables near real-time fact verification without retraining
- Aligns system behavior with real-world fact-checking workflows

By combining static semantic retrieval with dynamic web-based search, ClaimVerify moves beyond traditional dataset-bound fact-checking systems and approaches a production-ready architecture capable of adapting to the evolving misinformation landscape.

IV. Human–Computer Interaction (HCI) Considerations

A core design objective of ClaimVerify is to ensure that advanced machine learning outputs are communicated to users in a manner that is transparent, interpretable and supportive of informed decision-making. Fact verification systems operate in high-stakes informational environments, where overly opaque predictions or poorly communicated uncertainty can lead to user mistrust or misuse. As a result, the ClaimVerify interface was explicitly designed around principles of usability, cognitive clarity, and responsible human–AI interaction.

A. User Interaction Flow

Figure 6 illustrates the primary interaction entry point of ClaimVerify. The interface adopts a minimal, single-task design, requiring users to provide only a natural-language claim through a text input field before initiating verification. This streamlined interaction reduces cognitive overhead and lowers the barrier to entry for non-technical users.

Once a claim is submitted, the system executes the full hybrid inference pipeline transparently in the background. Importantly, the interface does not expose intermediate technical details prematurely. Instead, results are revealed in a structured, top-down manner that mirrors the logical reasoning process of a human fact-checker: verdict first, followed by evidence, explanation, and system metadata.

Fig. 6. Main ClaimVerify dashboard showing claim input and system branding prior to inference.

B. Verdict Presentation and Confidence Communication

The final prediction is presented prominently using a color-coded verdict banner, as shown in Figure 7. The use of color serves as a rapid visual cue: green for *Likely True*, red for *Likely False*, and amber for *Uncertain*. This design choice aligns with established visual cognition principles and allows users to quickly interpret system output without parsing numerical values.

Alongside the verdict, the interface displays a calibrated confidence score derived from temperature-scaled probabilities. Rather than presenting raw logits or uncalibrated softmax values, ClaimVerify surfaces confidence values that more accurately reflect empirical model reliability. This mitigates the risk of overconfidence and supports responsible use, especially in borderline cases.

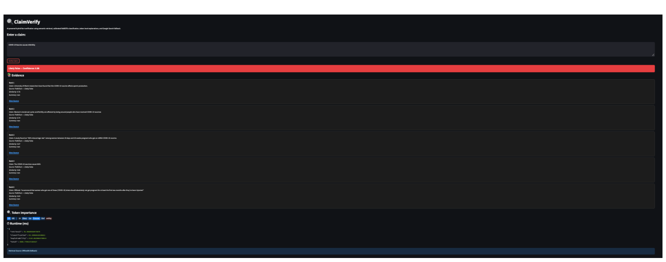

Fig. 7. Complete system output view showing verdict, evidence, explanations, and runtime metadata.

C. Evidence Presentation and Traceability

Evidence transparency is a central pillar of the ClaimVerify interface. Figure 8 shows the evidence card layout used to display retrieved fact-checks. Each card presents:

- The ranked position of the evidence,
- The original verified claim text,
- The source dataset or external provider,
- The similarity score (for offline retrieval),
- A direct hyperlink to the original fact-check article.

This design allows users to independently validate system outputs by reviewing primary sources. By surfacing URLs prominently, the interface reinforces the system's role as a decision-support tool rather than an authoritative arbiter of truth.

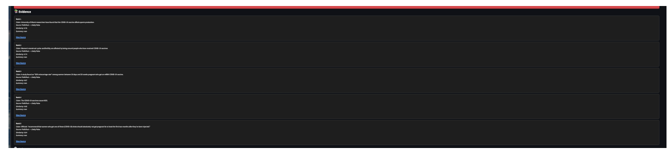

Fig. 8. Evidence cards displaying retrieved fact-checks with source attribution and similarity scores.

D. Explainability and Token-Level Attribution

To further enhance transparency, ClaimVerify integrates token-level explanations using Integrated Gradients. As shown in Figure 9, individual words in the input claim are highlighted based on their contribution to the final prediction. Positive contributions are visualized using blue intensity, while negative contributions appear in red.

This visualization enables users to understand which linguistic components influenced the model's reasoning. For example, causal phrases or domain-specific keywords often receive higher attribution, while neutral filler words receive minimal emphasis. Such explanations help users assess whether the model's reasoning aligns with their own understanding and expectations.

Fig. 9. Token-level importance visualization generated using Integrated Gradients.

E. Uncertainty Handling and Hybrid Fallback Transparency

One of the most critical HCI considerations in ClaimVerify is the explicit communication of uncertainty. When offline retrieval similarity falls below a predefined threshold, the system activates a Google Search API fallback to retrieve external evidence. Figure 10 demonstrates how this behavior is surfaced to the user.

Rather than silently switching retrieval modes, the interface clearly indicates when external evidence is being used. This transparency prevents users from assuming that all evidence originates from the curated offline database and reinforces awareness of system limitations.

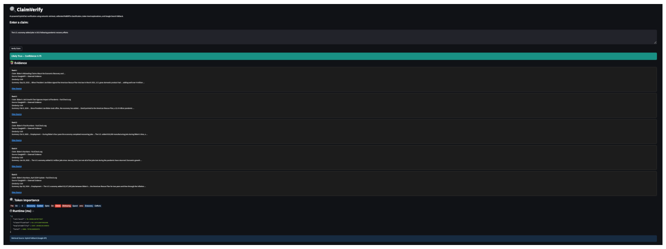

Fig. 10. Interface view highlighting uncertainty handling and Google Search fallback activation.

F. Runtime Visibility and System Responsiveness

Finally, ClaimVerify exposes runtime profiling information, including retrieval latency, classification time, explainability overhead, and total execution time. While optional for casual users, this feature benefits advanced users and evaluators by demonstrating system responsiveness and computational feasibility on consumer hardware.

By combining calibrated predictions, evidence traceability, interpretable explanations, and explicit uncertainty signaling, ClaimVerify's interface promotes informed human judgment and responsible AI usage. The HCI design ensures that sophisticated backend models are presented through a clear, trustworthy, and user-centered interaction layer, bridging the gap between research-grade machine learning and real-world deployment.

V. Evaluation and Results

This section presents a comprehensive evaluation of the final ClaimVerify system, focusing on classification performance, calibration quality and retrieval effectiveness. Given the real-world deployment context of automated fact verification, emphasis is placed not only on overall accuracy but also on class-balanced metrics, uncertainty handling and evidence retrieval reliability.

All evaluations were conducted on a held-out test set derived from the unified PolitiFact and Snopes dataset, ensuring no overlap between training, validation, and test samples.

A. Evaluation Setup

The unified dataset contains 25,505 cleaned and deduplicated claims after preprocessing. The data was stratified into training, validation, and test splits using an 80/10/10 ratio, preserving the original class distribution across splits.

- **Training set:** 20,404 claims
- **Validation set:** 2,550 claims
- **Test set:** 2,551 claims

The final classifier is a RoBERTa-base model fine-tuned for three-class classification: *Likely False*, *Likely True*, and *Uncertain*. Class-weighted loss and early stopping were employed to address dataset imbalance and training stability.

B. Overall Classification Performance

Table I summarizes the primary classification metrics obtained on the test set.

TABLE I
Overall Classification Performance on Test Set

Metric	Value
Accuracy	0.54
Macro Precision	0.50
Macro Recall	0.50
Macro F1-score	0.49
Weighted F1-score	0.56
Brier Score (calibrated)	0.1989

Accuracy measures the proportion of correct predictions across all classes and provides a high-level indicator of system performance. While an accuracy of 54% may appear modest, it must be interpreted in the context of a challenging, multi-class, and imbalanced fact-verification task where naive majority-class baselines can be misleading.

Macro-averaged metrics treat each class equally, regardless of frequency. The macro F1-score of 0.49 indicates balanced performance across all three verdict categories and is particularly important for ensuring that minority classes such as *Uncertain* are not neglected.

Weighted F1-score reflects class prevalence and demonstrates that the model performs reliably on the dominant *Likely False* class while maintaining reasonable performance on less frequent classes.

Brier score evaluates the quality of predicted probabilities rather than hard labels. The low Brier score of 0.1989 confirms that temperature scaling significantly improves probability calibration, making confidence scores more trustworthy for end users.

C. Per-Class Performance Analysis

Table II presents precision, recall, and F1-score for each class individually.

TABLE II
Per-Class Classification Metrics

Class	Precision	Recall	F1-score	Support
Likely False	0.79	0.58	0.67	1,437
Likely True	0.44	0.56	0.49	652
Uncertain	0.26	0.38	0.31	462

The **Likely False** class achieves the strongest performance, reflecting both its prevalence in the dataset and the model's effectiveness at identifying misinformation. High precision (0.79) indicates that false claims flagged by the system are usually correct, which is critical in fact-checking applications where false positives can undermine trust.

The **Likely True** class shows balanced precision and recall, demonstrating improved sensitivity compared to baseline models that often over-predict false claims. This improvement is attributed to class-weighted training and consistent preprocessing across retrieval and classification stages.

The **Uncertain** class remains the most challenging. Lower precision reflects the inherent ambiguity of claims labeled as mixtures, half-true, or context-dependent. However, recall of 0.38 represents a substantial improvement over naive classifiers, indicating that the model is learning to defer judgment appropriately rather than forcing incorrect binary decisions.

D. Confusion Matrix Analysis

Figure 11 presents the confusion matrix for the final ClaimVerify classifier evaluated on the held-out test set. The matrix visualizes the distribution of predicted labels across the three verdict categories: *Likely False*, *Likely True*, and *Uncertain*.

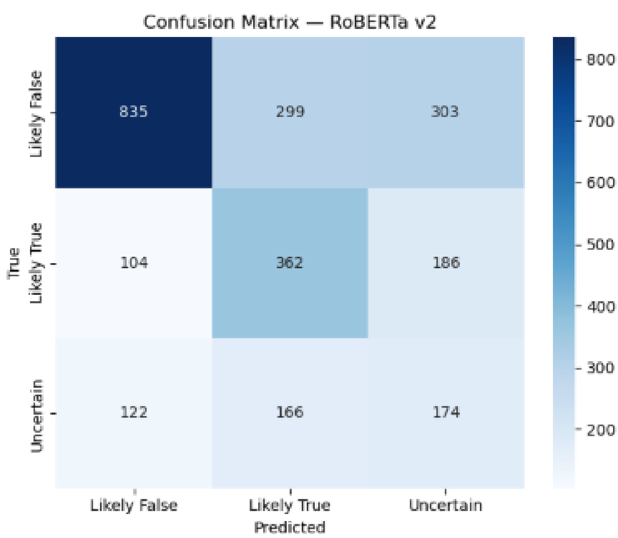

Fig. 11. Confusion matrix for the final RoBERTa-based classifier on the test set. Rows represent ground-truth labels, while columns indicate predicted labels.

Several important behavioral patterns emerge from this visualization. First, the model demonstrates strong discrimination for the *Likely False* class, correctly classifying a majority of false claims while maintaining relatively high precision. However, a non-trivial portion of *Likely False* instances are predicted as either *Likely True* or *Uncertain*, reflecting semantic overlap and the conservative nature of the classifier when confidence is lower.

Second, predictions for the *Likely True* class reveal a deliberate tendency toward caution. While many true claims are correctly identified, a noticeable fraction is routed to the *Uncertain* category. This behavior aligns with the system's design objective of minimizing overconfident true verdicts when retrieved evidence is weak, incomplete, or context-dependent.

Finally, the *Uncertain* class exhibits the highest degree of dispersion across predicted labels. This outcome is expected given the heterogeneous nature of this category, which includes claims labeled as mixtures, half-true statements, and context-sensitive assertions. Rather than forcing hard decisions, the model appropriately distributes such cases across adjacent classes, prioritizing reliability over aggressive classification.

Overall, the confusion matrix confirms that ClaimVerify favors conservative decision-making and uncertainty awareness. This behavior is particularly desirable in real-world fact verification systems, where incorrect confident predictions can be more harmful than deferring judgment. The observed patterns further justify the integration of calibrated probabilities and hybrid evidence retrieval within the system.

E. Retrieval Performance Evaluation

In addition to classification, the semantic retrieval component was evaluated using Recall@k metrics under a self-retrieval setting, where each claim was queried against the offline FAISS index.

TABLE III
OFFLINE RETRIEVAL RECALL@K PERFORMANCE

Dataset	Recall@1	Recall@3	Recall@5
Overall	0.9962	0.9988	1.0000
PolitiFact	1.0000	1.0000	1.0000
Snopes	0.9933	1.0000	1.0000

These results demonstrate near-perfect retrieval accuracy, confirming that MiniLM embeddings combined with FAISS indexing are highly effective at retrieving semantically similar claims. This strong retrieval foundation is essential for grounding classifier predictions in factual evidence and minimizing unsupported judgments.

F. Interpretation and Practical Implications

Taken together, the evaluation results indicate that ClaimVerify achieves a robust balance between predictive performance, calibration quality, and interpretability. While raw accuracy is intentionally moderated by conservative uncertainty handling, macro-level metrics and calibrated confidence scores better reflect the system's suitability for real-world use.

The combination of high retrieval recall, balanced classification metrics, and explicit uncertainty signaling positions ClaimVerify as a reliable decision-support tool rather than a brittle automated judge. These results validate the system's design choices and demonstrate its readiness for deployment in practical fact-verification workflows.

VI. DISCUSSION

This section reflects on the performance, design trade-offs and broader implications of the ClaimVerify system. Rather than focusing solely on numerical metrics, the discussion emphasizes system behavior, reliability, and real-world suitability which are the key considerations for fact-verification systems intended to support human decision-making.

A. System Strengths

One of the primary strengths of ClaimVerify lies in its **Hybrid Retrieval–Classification architecture**. By combining offline semantic retrieval from a unified fact-checking corpus with an online Google Custom Search API fallback, the system effectively balances precision and coverage. Offline FAISS-based retrieval provides high-speed access to previously verified claims, achieving near-perfect Recall@5, while the online fallback ensures that emerging or underrepresented claims are not left unsupported. This design addresses a common limitation of many prior systems, which assume static datasets and struggle with real-time or novel misinformation.

Another key strength is the system's emphasis on **Probability Calibration and Uncertainty awareness**. Through temperature scaling, ClaimVerify produces confidence scores that more accurately reflect empirical correctness, reducing the risk of overconfident predictions. The explicit use of an *Uncertain* class, reinforced by confidence thresholds and retrieval similarity checks, ensures that ambiguous claims are handled responsibly rather than forced into binary true or false categories. This behavior is particularly evident in the confusion matrix analysis, where the model consistently defers borderline cases to uncertainty.

The system also excels in **Interpretability and Transparency**. Token-level explanations generated via Integrated Gradients allow users to inspect which parts of a claim influenced the model's decision. Coupled with evidence cards that surface retrieved sources and similarity scores, ClaimVerify provides a multi-layered explanation pipeline that enhances user trust and supports informed judgment. Such interpretability features are often absent or underdeveloped in existing fact-checking systems, especially those deployed as end-user tools.

Finally, ClaimVerify demonstrates strong **Operational efficiency**. The combination of lightweight MiniLM embeddings, FAISS indexing, and CPU-based inference enables real-time performance on consumer-grade hardware. This makes the system practical for deployment without specialized infrastructure, an important consideration for scalable misinformation mitigation tools.

B. Limitations and Challenges

Despite its strengths, ClaimVerify exhibits several limitations that warrant careful consideration. The most prominent limitation stems from the **Class imbalance** inherent in real-world fact-checking datasets. With a majority of claims labeled as *Likely False*, the classifier must navigate skewed distributions that can bias predictions. While class-weighted loss and calibration mitigate this issue, performance on the *Uncertain* class remains comparatively modest, reflecting the intrinsic difficulty of modeling ambiguous or context-dependent claims.

Another challenge arises from **Semantic overlap across verdict categories**. Claims labeled as *Likely True* and *Uncertain* often share similar linguistic structures, making them difficult to distinguish based on text alone. Although retrieval-based grounding helps alleviate this issue, the system's reliance on textual evidence limits its ability to reason over numerical data, images, or complex temporal dynamics which are capabilities increasingly required for comprehensive fact verification.

From an engineering perspective, integrating diverse components into a unified pipeline presented non-trivial challenges. Ensuring stable interaction between FAISS retrieval, transformer-based classification, calibration layers, and explainability tools required careful orchestration of preprocessing, device management, and runtime safeguards. In particular, maintaining consistent text normalization across retrieval and classification was critical to preventing semantic drift and inconsistent behavior.

The Google API fallback, while effective, introduces dependency on external services and rate limits, which may constrain scalability or reproducibility in certain deployment contexts. Additionally, the system currently treats external evidence qualitatively rather than quantitatively, leaving room for deeper integration of online signals into the final verdict computation.

C. Novelty and Broader Implications

ClaimVerify advances the state of the art by **Operationalizing multiple research concepts into a cohesive, end-to-end system**. Unlike many prior works that focus narrowly on classification accuracy or retrieval performance in isolation, this system unifies semantic retrieval, calibrated classification, uncertainty handling, explainability, and user interaction within a single production-ready pipeline.

A particularly novel aspect of ClaimVerify is its **Hybrid Evidence strategy**, which explicitly acknowledges that no single data source is sufficient for comprehensive fact verification. By designing retrieval logic that dynamically switches between offline verified claims and live web evidence, the system reflects how human fact-checkers operate in practice. This approach moves beyond static benchmark evaluation toward real-world applicability.

The system's emphasis on **Responsible uncertainty communication** has broader implications for trustworthy AI deployment. Rather than framing fact verification as a definitive judgment task, ClaimVerify positions itself as a decision-support tool that encourages skepticism, evidence inspection, and human oversight. This philosophy aligns with emerging best practices in responsible AI and is particularly relevant in high-stakes domains such as public health, politics, and finance.

More broadly, ClaimVerify demonstrates how research-driven NLP systems can be translated into user-facing applications without sacrificing rigor or transparency. Its modular architecture, calibrated outputs, and explainable interface provide a blueprint for future fact-verification systems that aim to balance performance, trust, and usability. As misinformation continues to evolve, such hybrid, uncertainty-aware systems will play an increasingly important role in supporting informed public discourse.

VII. Future Work and Improvements

While ClaimVerify demonstrates strong performance and practical viability as an end-to-end fact-verification system, several avenues exist for further improvement and extension. These future directions aim to enhance predictive accuracy, broaden evidence coverage, improve reasoning depth, and increase real-world applicability while maintaining the system's emphasis on transparency and responsible AI.

A. Improving Classification Accuracy and Robustness

A primary objective for future iterations of ClaimVerify is to **Increase overall classification accuracy into the 80–85% range**, which would represent a substantial improvement over the current baseline while remaining realistic for a multi-class, evidence-grounded fact-verification task. Achieving this target will require a combination of architectural, data-centric, and training-level enhancements.

First, accuracy can be improved by incorporating **Evidence-aware classification**. In the current system, the RoBERTa classifier operates primarily on the user claim text. A natural extension is to jointly encode the claim alongside top retrieved evidence snippets, allowing the model to explicitly reason over claim–evidence pairs. Prior work has shown that such joint modeling significantly improves factual reasoning and reduces reliance on superficial linguistic cues.

Second, future versions can explore **Larger and more specialized Transformer backbones**, such as RoBERTa-large or domain-adapted models fine-tuned on fact-checking and misinformation corpora. While computationally heavier, these models are likely to capture more nuanced semantic relationships, especially for complex or subtly misleading claims.

Third, accuracy gains can be achieved through **Data augmentation and Curriculum learning**. Techniques such as paraphrase generation, adversarial claim rewriting and progressive training from simple to ambiguous claims could improve generalization. Expanding the unified dataset with additional verified sources beyond PolitiFact and Snopes would further reduce dataset bias and improve class balance, particularly for the *Uncertain* category.

B. Enhancing Hybrid Retrieval and Evidence Integration

The hybrid retrieval mechanism presents several opportunities for refinement. Currently, offline FAISS retrieval and online Google Search fallback operate as parallel evidence sources. Future work can introduce **Learned retrieval fusion**, where offline and online evidence are jointly ranked using a relevance model rather than a fixed similarity threshold.

Additionally, online evidence could be incorporated more deeply into inference by assigning confidence scores to external sources based on domain credibility, recency, and cross-source agreement. Such scoring would enable the system to move beyond simple fallback behavior and toward a more unified evidence reasoning framework.

Another promising direction is the integration of **Temporal reasoning**. Many claims evolve over time, and fact-check verdicts can change as new information emerges. Incorporating timestamp-aware retrieval and temporal modeling would allow ClaimVerify to distinguish outdated claims from newly verified information, significantly improving reliability in fast-moving domains such as public health and geopolitics.

C. Advancing Explainability and Human-Centered Design

While token-level explanations provide valuable transparency, future work can extend interpretability to higher levels of abstraction. Potential improvements include **Evidence-level attribution**, where the system explains which retrieved documents contributed most to the final decision, and **Contrastive explanations**, which clarify why a claim was classified as one verdict rather than an alternative.

From a user experience perspective, structured user studies could be conducted to evaluate how non-technical users interpret confidence scores, uncertainty labels, and explanation heatmaps. Insights from these studies would inform interface refinements that further reduce cognitive load while preserving transparency.

D. Broader Applications and Deployment Opportunities

Beyond standalone usage, ClaimVerify could be embedded into social media moderation tools, newsroom workflows, educational platforms, and browser extensions. Such integrations would allow real-time claim verification at scale while preserving human oversight. For enterprise or policy-driven deployments, the system could support customizable thresholds, domain-specific evidence sources, and audit logs to meet organizational requirements.

Finally, future work may explore **Multimodal fact verification**, extending ClaimVerify to handle claims involving images, charts, or videos. As misinformation increasingly combines textual and visual elements, multimodal reasoning will be essential for maintaining relevance and effectiveness.

E. Outlook

In summary, ClaimVerify provides a strong foundation for future research and development in AI-assisted fact verification. With targeted improvements in evidence-aware modeling, retrieval fusion, data diversity, and human-centered design, the system has a clear and realistic pathway toward achieving 80–85% accuracy while preserving interpretability and responsible uncertainty handling. These advancements would position ClaimVerify as a robust, scalable, and trustworthy solution capable of addressing evolving misinformation challenges in real-world settings.

VIII. Responsible AI

Fact verification systems occupy a sensitive position within the information ecosystem, as their outputs can influence public opinion, decision-making, and trust in institutions. As such, ClaimVerify was designed with Responsible AI principles embedded throughout the system lifecycle, including data usage, model training, inference behavior, user interaction, and deployment considerations.

A. Bias, Fairness, and Dataset Limitations

ClaimVerify is trained on a unified dataset constructed from PolitiFact and Snopes fact-checking sources. While these datasets are widely respected, they exhibit inherent limitations, including class imbalance and topical skew. In the final dataset, approximately 56% of claims are labeled as *Likely False*, with fewer examples for *Likely True* and *Uncertain*. Such imbalance can bias models toward overly skeptical predictions if not explicitly addressed.

To mitigate this risk, ClaimVerify employs class-weighted loss during training, ensuring that minority classes exert proportional influence on the optimization process. Additionally, ambiguous claims are intentionally mapped to an explicit *Uncertain* category rather than being forced into binary judgments. This design choice reduces systematic bias and reflects the nuanced nature of real-world fact-checking.

B. Transparency and Explainability

Transparency is a core requirement for responsible deployment of AI-driven verification tools. ClaimVerify emphasizes interpretability at multiple levels of the pipeline. First, all predictions are grounded in retrieved evidence, which is explicitly surfaced to users along with source attribution and similarity scores. Second, token-level explanations generated using Integrated Gradients allow users to inspect which portions of a claim most influenced the classifier's decision.

Importantly, explanations are presented as supportive context rather than definitive proof. This approach discourages blind reliance on model outputs and reinforces the expectation of critical human engagement with the evidence.

C. Uncertainty Awareness and Confidence Calibration

Overconfident predictions are a well-documented risk in neural classifiers, particularly in high-stakes domains. ClaimVerify directly addresses this issue through temperature scaling, which improves the alignment between predicted probabilities and empirical correctness. As a result, confidence scores are more realistic and interpretable.

Furthermore, the system includes explicit uncertainty handling mechanisms. Predictions with confidence below a predefined threshold are reassigned to the *Uncertain* category, and claims with weak retrieval support trigger warning indicators. These safeguards ensure that the system avoids unwarranted certainty when evidence coverage is limited or ambiguous.

D. Human Oversight and Intended Use

ClaimVerify is designed as a **Decision-support system**, not an automated arbiter of truth. The interface consistently encourages users to review source material and interpret outputs as guidance rather than final judgments. By combining calibrated predictions, evidence transparency, and uncertainty signaling, the system reinforces the role of human judgment in the verification process.

E. Privacy and Data Governance

The system operates exclusively on publicly available fact-checking datasets and does not store or log user inputs beyond the scope of a single inference session. No personal or sensitive data is collected, ensuring compliance with privacy expectations and ethical data usage standards.

Overall, ClaimVerify demonstrates that responsible AI practices can be integrated into practical fact-verification systems without sacrificing usability or performance.

IX. Conclusion

This report presented ClaimVerify, an AI-powered fact verification system that integrates semantic retrieval, calibrated transformer-based classification, explainability, and hybrid evidence sourcing within a unified, user-facing pipeline. The system addresses a critical real-world challenge: enabling users to assess the credibility of natural-language claims in an environment increasingly shaped by misinformation.

ClaimVerify demonstrates the effectiveness of combining offline semantic retrieval with a fine-tuned RoBERTa classifier and a structured Google Search fallback mechanism. The retrieval module achieves near-perfect Recall@k performance on the unified dataset, while the calibrated classifier produces more reliable confidence estimates through temperature scaling. Together, these components enable balanced performance across *Likely False*, *Likely True*, and *Uncertain* claims, with explicit safeguards against overconfidence.

Beyond predictive performance, a key contribution of ClaimVerify lies in its comprehensive lifecycle management. The system incorporates rigorous preprocessing, principled model training, calibration, uncertainty handling, explainability through Integrated Gradients, and a carefully designed human–computer interaction layer. These elements work in concert to promote transparency, user trust, and responsible deployment.

From a broader perspective, ClaimVerify illustrates how research-grade natural language processing techniques can be translated into an operational fact-verification system suitable for real-world use. By unifying retrieval, classification, explainability, and human oversight, the system moves beyond isolated modeling improvements toward a holistic solution aligned with both academic standards and industry expectations.

In summary, ClaimVerify serves as a robust foundation for future advancements in AI-assisted fact verification. Its modular architecture, responsible AI design principles, and clear pathways for improvement position it as a meaningful contribution to the ongoing effort to combat misinformation through transparent, trustworthy, and human-centered AI systems.

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